1. Infectious Diseases

1.1. Infectious Diseases

- Disease:
  - Poor health or illness causes reduced effectiveness of functions; has a set of symptoms, can be poor physical, mental, and social well-being
- Infectious diseases:
  - Transmissible diseases caused by a pathogen which cause a reduced effectiveness of function

Disease 1: Cholera

Item	Description
Pathogen	Vibrio cholerae
Type of pathogen	Bacteria
Transmission	Faecal-oral route
Site of action	Walls of small intestine
Method of action	Bacteria secrete choleragen toxin, which binds to a complementary receptor on intestinal cells via endocytosis, loss of Na+ and Cl- from cells; water moves out of the blood
Symptoms	Severe diarrhoea, severe dehydration, loss of water and salts, fatigue
Treatment	Oral rehydration therapy (ORT)
Prevention	Proper sewage treatment, chlorinated water, drinking bottled waters, no effective vaccine due to mutation, and several strains
Global Distribution	Outbreaks follow natural disasters due to lack of proper sanitation, poor hygiene, and poor living conditions.
Diagnosis	Dipstick test using rectal swabs.

Disease 2: Malaria

Item	Description
Pathogen	Plasmodium falciparum Plasmodium vivax Plasmodium ovale Plasmodium malariae
Type of pathogen	Protoctist
Transmission	Vector female Anopheles mosquito \n Mosquito takes a blood meal, parasite enters the host
Site of action	RBCs, liver, brain
Method of action	The parasite enters the bloodstream and matures in liver cells, enters RBCs, and divides in RBCs, causing them to lyse, infect other RBCs
Symptoms	Fever, anaemia, enlarged spleen
Diagnosis	Dipstick test, microscopical analysis of blood
Treatment	Prophylactic antimalarial drugs; Artemisinin-based combination therapy (ACT)
Prevention	Use preventative drugs to reduce the number of mosquitoes: Spray insecticides and spread oil over the water surface to prevent breeding. Breeding fish that feed on larvae Spray Bacillus thuringiensis bacteria to kill larvaePrevention of bites: Use mosquito nets soaked in insecticidesRepellents No vaccination available
Global Distribution	Endemic in tropical and sub-tropical areas: - vector breeds in hot and humid and needs stagnant water to reproduce - Plasmodium reproduces within mosquitoes at >20C
Other Problems	Drug-resistant Plasmodium, insecticide DDT-resistant mosquito, difficulty developing a vaccine, migration

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Disease 3: HIV/AIDS

Item	Description
Pathogen	Human immunodeficiency virus (HIV)
Type of pathogen	Virus (RNA retrovirus)
Transmission	Direct exchange of bodily fluids through semen/vaginal fluids during sexual intercourse, blood transfusions, mother-to-baby across placenta/breast milk
Site of action	T-helper lymphocytes
Method of action	Slow infection. Viral RNA and reverse transcriptase (RT) enter T-helper cells and convert RNA → DNA. Viral DNA is incorporated into host DNA, and cells express viral proteins, so T-helper cells cannot function.
Symptoms	AIDS (acquired immunodeficiency syndrome) and opportunistic infections e.g. TB, malaria, oral thrush, cancers
Diagnosis	Blood/saliva/urine test
Treatment	No vaccination, no cure. Drugs to slow down the onset of AIDS: Zidovudine inhibits RT and viral enzymes. Combination therapy
Prevention	Use condoms, femidoms, and dental dams. Provide HIV testing centers and contact tracing. Control mother-child transmission using drugs. High-income HIV+ women should avoid breastfeeding to reduce transmission. Low-income HIV+ women should breastfeed for passive immunity.

Disease 4: Tuberculosis

Item	Description
Pathogen	Mycobacterium tuberculosis Mycobacterium bovis
Type of pathogen	Bacteria
Transmission	M. tuberculosis: aerosol infection. The infected person sneezes, and the uninfected person breathes in the airborne droplets. M. bovis: from infected cattle e.g. unpasteurized milk
Site of action	Lungs
Method of action	Slow infection. The disease can stay dormant and become active later when the immune system is weakened e.g., by HIV.
Symptoms	Racking cough, shortness of breath, coughing blood
Diagnosis	Chest X-ray, microscopical examination of sputum
Treatment	Combination therapy of several antibiotics to prevent resistance. Through DOTS (direct observation treatment, short course)
Prevention	BCG vaccine, contact tracing, quarantine, pasteurise milk
Global Distribution	Endemic due to high rate of migration, HIV/AIDS prevalence, overcrowded living spaces
Other Problems	MDR-TB and XDR-TB require longer treatment with more expensive antibiotics

1.2. Antibiotics

- A substance derived from a living organism that is capable of killing or inhibiting the growth of a microorganism

How they work:

- Interfere with some aspect of growth or metabolism of the target bacterium, including,
  - Synthesis of bacterial cell walls
  - The activity of proteins in the cell surface membrane
  - Enzyme action
  - DNA and protein synthesis

How penicillin works:

- Penicillin inhibits the transpeptidase enzyme
- Stops formation of cross-links between peptidoglycan polymer in cell wall
- Autolysin continues making holes, but cross-links are not formed as penicillin inhibits transpeptidase enzyme
- These little holes allow the wall to stretch so that new peptidoglycan chains can link together
- The cell wall, therefore, becomes progressively weaker, and it bursts due to osmotic pressure
- Bacteria lyses

Antibiotics do not affect viruses because:

- Virus do not have peptidoglycan cell walls, they have a protein coat
- Virus do not have their metabolism
- Virus has very few organelles, so very few sites of action
- Virus live inside host cells, out of reach of antibiotics

DNA mutation:

- An existing gene within the bacterial genome changes spontaneously to give rise to a nucleotide sequence that codes for a slightly different protein that is not affected by the antibiotic

How resistance to antibiotics develops

- Overuse of antibiotics or misuse of antibiotics, e.g. taking antibiotics for viral infections, overprescribing for bacterial infections
- Not completing an antibiotic course
- The reservoir of bacteria remains
- Antibiotic acts as a selection pressure and induce a mutation in bacteria, which makes them resistant
- Mutation, e.g. protein produced has a changed binding site
- Bacteria with resistance survive, and only bacteria sensitive to antibiotics are killed (antibiotic acts as a selection pressure)
- Bacteria reproduce and pass on genes/alleles for resistance to offspring by:
  - Asexual reproduction (vertical transmission) or
  - Conjugation, which transfers resistance genes to the plasmid (horizontal transmission)

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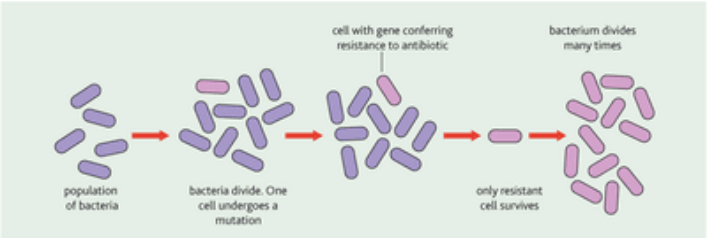
- Frequency of resistance alleles/gene increase in bacterial population

Selection pressure:

- Any phenomenon which alters the behaviour and fitness of living organisms within a given environment. It is the driving force of evolution and natural selection, e.g., natural disasters, predators, diseases, etc.



Reducing bacterial resistance:

- Use antibiotics only when appropriate and necessary; not prescribing them for viral infections
- Making sure that patients complete their course of medication
- Medicines should only be given if prescribed
- Narrow-spectrum drugs should be used rather than wide-spectrum drugs
- Reduce the use of antibiotics in drugs in agriculture
- Use of many antibiotics to destroy the pathogen to make sure it doesn't mutate

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